Night-time temperatures drive maize yield variability in the US Mid-South

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Complete List of Authors:	Nores Allende, Maximo ; University of Arkansas, Crop, Soil, and Environmental Sciences dos Santos, Caio; Iowa State University of Science and Technology College of Agriculture and Life Sciences, Agronomy Carlin, John; University of Arkansas Kelley, Jason; University of Arkansas Fayetteville Elli, Elvis; University of Arkansas System
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Crop season temperatures are expected to increase during the coming decades, which can reduce maize yields. Therefore, quantifying the impacts of temperature on maize yield is critical to minimize its negative affect. Our goal was to identify which weather variables were the most important ones affecting yields and how increases in night-time temperature affect yields. The results show that night-time temperature had the strongest negative effect on yields. For each 1Â°C increase in night-time temperature, yields decreased 3.7%. Moreover, cumulative solar radiation had the strongest positive effect, meaning that more solar radiation availability during the cycle increased yields. The findings can help to adjust practices such as planting date and hybrid selection to minimize the negative impact of high night-time temperatures and increase maize yields.

Night-time temperatures drive maize yield variability in the US Mid-South

Maximo Nores Allende[1], Caio dos Santos[2], John Carlyn[1], Jason Kelley[1], Elvis F. Elli[1]

[1] Department of Crop, Soil, and Environmental Sciences, University of Arkansas System Division of Agriculture, Fayetteville, Arkansas, USA.

[2] Department of Agronomy, Iowa State University, Ames, Iowa, USA.

*Correspondence eelli@uark.edu (E.F.Elli)

Abbreviations: NT, nighttime temperature; MinT, minimum temperature; MaxT, maximum temperature; CumRad, cumulative radiation; SD, standard deviation.

- Minimum temperature was the most important variable determining maize yields.
- For every 1°C increase in minimum temperature, yield decreased by 3.7%.
- Mixed-effect models are powerful in predicting maize yield using weather information.

Abstract

High night-time temperatures (NT) are an emerging threat to maize, and understanding the influence of NT on yields is critical for quantifying climate-related yield penalties. The objectives of the study were to i) identify the impacts of weather variables on maize yield using mixed-effects models, and ii) run a sensitivity analysis to identify the implications of altered NT on maize yields in the context of climate change. A comprehensive multi-environmental dataset comprising maize grain yield across 64 site-years and environmental weather variables was used to fit the models. Night-time temperature was the most important weather variable driving yields. For every 1°C increase in NT during the June-July period from the baseline scenario, yields declined 3.7% (500 kg ha^{-1}). The findings can refine management practices, including planting date and hybrid selection, to minimize yield penalties from NT, and enhance maize yield predictions by using NT as a predictor.

1. Introduction:

Deepening the understanding of the environmental factors driving maize yields is crucial to maximizing grain production globally, particularly as the global population continues to grow. Increasing population growth rates suggest that food demand will increase by 58-59% by 2050 (Valin et al., 2014). To meet the projected maize demands for the year 2050, global maize yields must increase at a rate of 2.4% per year (Ray et al., 2013). Expanding the knowledge of the effect of environmental conditions on maize yields is crucial to achieve this goal.

Even though maize has substantial resilience to high temperatures (Rotundo et al., 2019), heat stress can lead to severe yield loss during certain reproductive stages (Cirilo & Andrade, 1994b, 1994a; Jagadish, 2020; Lohani et al., 2020; Prasad et al., 2015), especially during the critical period, centered roughly 30 days around flowering. Numerous researchers have described the correlation between final yield and kernel set and highlighted the importance of environmental conditions during tasseling and silking on the final kernel set (Cirilo & Andrade, 1994a, 1994b; Otegui et al., 1995; Otegui & Bonhomme, 1998; Rattalino Edreira & Otegui, 2012). Surprisingly, the effect of night-time temperature (NT) on crops has been understudied compared to other environmental factors (Giménez et al., 2025; Serrago et al., 2025), even though it is certain that it has a substantial effect on grain yields (Impa et al., 2021).

Although often coinciding, high daytime temperature and high NT have different effects on yield (Giménez et al., 2025; Wang et al., 2020). Extreme heat events directly affect growth, photosynthesis, and kernel development, while high NT during flowering accelerates development, and increases night respiration, lowering kernel number. Increased development rates diminish resource accumulation and lead to the premature senescence of the leaves (García et al., 2015; Impa et al., 2021; Kettler et al., 2022). Previous studies demonstrated that the increase in minimum temperature (MinT), which is highly correlated with NT, can reduce grain yields by up to 10% for every 1°C increase, depending on the phenological stage and growing conditions (García et al., 2015; Impa et al., 2021; Kettler et al., 2022). Warmer nights around flowering and silking reduced grain yield by ~24% and kernel number by ~25% through accelerated development rates and kernel abortions (Kettler et al., 2022; Wang et al., 2020). If heating is extended into the grain filling period, grain weight may also be affected (García et al., 2015; Kettler et al., 2024).

Even though previous research has investigated the effect of high NT on maize yields (Cantarero et al., 1999; Kettler et al., 2022, 2024; Wang et al., 2020) and in other cereal crops (García et al., 2015; Impa et al., 2021), evidence suggests that the isolated effect of NT on maize yield remains understudied (Kettler et al., 2022, 2024; Serrago et al., 2025; Wang et al., 2020). Our study advances previous theoretical and controlled environment studies (García et al., 2015; Impa et al., 2021; Wang et al., 2020) by quantifying the impacts of high NT on maize yields using 15 years of field experimentation, totaling 64 site-years spread across the US Mid-South. The objectives of the study were to: i) identify the impacts of different weather variables on maize yield using mixed-effects models, and ii) conduct a sensitivity analysis using the fitted model and identify the implications of altered NT on maize yields in the context of climate change.

2. MATERIALS AND METHODS:

2.1. Environments and Experimental Design

This study used data from a multi-environmental trial experiment conducted by the Variety Testing Programs from the University of Arkansas System Division of Agriculture

(Carlin et al., 2026). Over a 15-year period, from 2010 to 2024, a total of 798 commercial maize hybrids were evaluated in seven separate locations across Arkansas, US. Relative maturity ranged from 103 to 121 days. Cumulative monthly rainfall (Figure S1) and MinT (Figure S2) across locations ranged from 10 mm to 170 mm and from 9°C to 23.8°C, respectively. The locations across Arkansas included five University research stations—Keiser, Marianna, Stuttgart, Rohwer, and Harrisburg, and a commercial farmer-managed field, Des-Arc (Figure S3). Experiments followed a complete randomized block design with four replications, and all fields were furrow irrigated. Final grain yields (kg ha^{-1} at 0% moisture) were obtained at physiological maturity (R6) from two rows of 7.62 m using a small plot combine. Large yield variability was observed across sites, ranging from 7800 kg ha^{-1} to 19700 kg ha^{-1} (Table S1).

2.2. Weather data

Environmental information was obtained from NASA POWER (Stackhouse et al., 2015) and weather stations located at the research stations. The complete set of retrieved weather variables included mean MaxT (°C), mean MinT (°C), days above 35°C (n), vapor pressure deficit (kPa), cumulative radiation (CumRad) (MJ m^{-2} day^{-1}) and cumulative rain (mm). The average MinT, the average vapor pressure deficit (VPD), and days above 35°C included information for the June-July period of each site-year. The June-July period was selected because, according to USDA, NASS (2025), during the past 12 years, 100% of the maize area in Arkansas reached the critical period between early June to late July, making this period extremely relevant for yield definition. The locations capture different production situations in terms of weather and soil conditions. June-July mean temperatures ranged from 24.8°C to 31.1°C across the 64-site-years, while minimum temperatures ranged from 20.1°C to 24.6°C with increasing trends in maximum, minimum and mean temperature over that past 40 years in

Arkansas (Figure S4). Soil series ranged from moderately well-drained silt loams (Calhoun, Calloway) in Marianna and Des-Arc to poorly drained soils in Keiser and Rohwer (Sharkey and Sharkey-Desha).

2.3 Statistical analysis

To analyze the importance and magnitude of the different predictor variables of maize grain yield and to account for the multi-level structure of the data across years and locations, mixed-effect models were used (Gambin et al., 2025; Malosetti et al., 2013; Smith et al., 2005). Effects were considered as fixed when the effect of each of the levels of the factor was of interest (Gelman, 2005). Fixed effects included weather variables, while random effects included variables that represented sources of variability, but those specific effects were not the focus of this study (Location and Hybrid). Weather variables were standardized (mean = 0, standard deviation (SD) = 1) to enhance comparability of the effect size of the predictors and improve interpretability of the results. Model assumptions were verified by checking the residuals' normality and homogeneity of variance. Heteroscedasticity was addressed by allowing the residual variance to change across locations. Model selection was performed by fitting alternative models including different temperature variables and comparing the model's performance. The model featuring the lowest AIC and RMSE was selected. Variance Inflation Factor (VIF) values from the model were retrieved to check for multicollinearity among predictors, following Kim's (2019) thresholds. A significance level (α) of 0.05 was used.

2.4 Sensitivity Analysis

To evaluate how sensitive maize yield is to NT increases and the robustness of the model, a sensitivity analysis was conducted. The objective was to quantify how the predicted yield responds to shifts in MinT while holding the remaining covariates constant. A set of scenarios was generated by adding fixed increments relative to the MinT baseline of each environment. Specifically, scenarios were created representing increases of +1, +2, +3, and +4°C in MinT of each site to make yield predictions using the mixed-effects model. The model's predictive performance was assessed using leave-one-location-out cross-validation (RMSE) to assess model performance in unseen environments (Buntaran et al., 2021; Tadese et al., 2024). The sensitivity analysis allowed the comprehension of the effect of MinT on yield under hypothetical warming conditions, following projections of Elli et al. (2022) and McBride et al. (2021).

3. RESULTS:

The mixed-effects model using MinT had the lowest AIC and showed a good performance in fitting the data, with an RMSE of 1130 kg ha^{-1}. Model's VIF values indicated no evidence of concerning multicollinearity among the predictors (the highest VIF was 2.34). The model ranked the average MinT during the June-July period as the variable with the highest weight when determining maize yields in the Mid-South US (Figure 1). For every 1 SD increase in MinT, yields decreased 0.54 SD, meaning that higher values of this variable were associated with yield penalties. Since the flowering period and the beginning of the grain filling period of every site-year fall within the June-July period, the increased transpiration and the shortened grain filling period explain the weight of the MinT as the most important for yield determination.

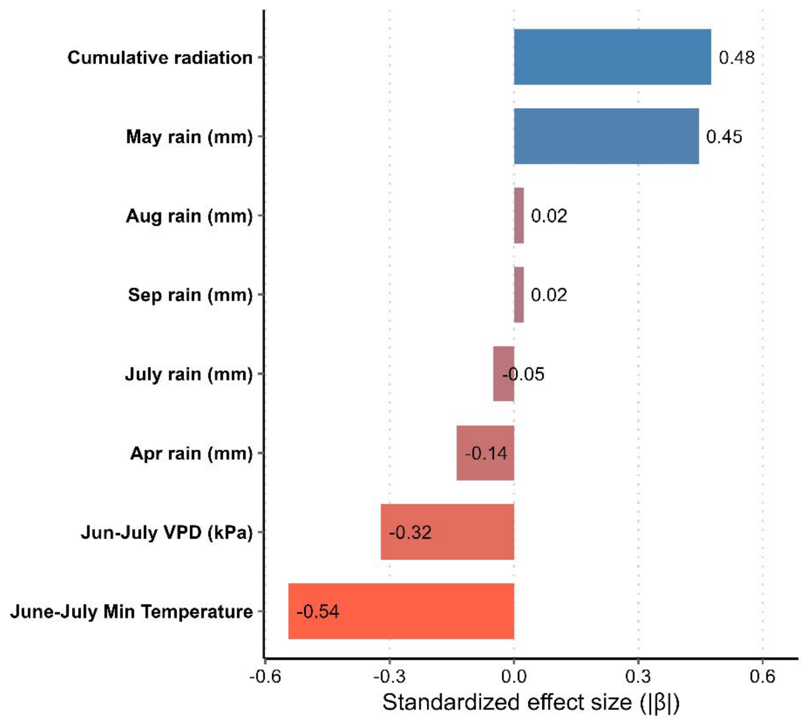

Figure 1. Relative importance of standardized climate predictors on determining maize yield across 64 site-years of data. Bars show standardized effect sizes with positive predictors in blue and negative in red.

Cumulative Radiation during the crop cycle had a positive effect ($\beta = 0.48$), indicating that a one SD increase in CumRad was associated with a 0.48 SD increase in yield. Surprisingly, cumulative rainfall in May was an important predictor of yields ($\beta = 0.45$), even though the experiments were furrow irrigated. The average VPD during the June-July period had a negative effect on yields ($\beta = -0.32$) as expected, indicating that an increase in VPD diminishes yields. The remaining predictors showed comparatively smaller effect sizes.

The mixed-effects model was used to create predictions on simulated environments where only NT was changed. The average observed yield was 13730 kg ha^{-1} and model-predicted yield under current NT conditions (13690 kg ha^{-1}) was used as baseline to calculate yield penalties in simulated scenarios. The cross-validated RMSE averaged 1630 kg ha^{-1}. While the magnitude of maize yield response to increases in NT varied with the environmental conditions across year-location combinations (Figure S5), an average yield penalty of 3.7% yield penalty for every 1°C increase in NT which represented 500 kg ha^{-1} grain yield of loss from the baseline yield (Figure 2). In the warmest scenario (+4°C in NT) the yield penalty was ~15%, which represented ~ 2000 kg ha^{-1}.

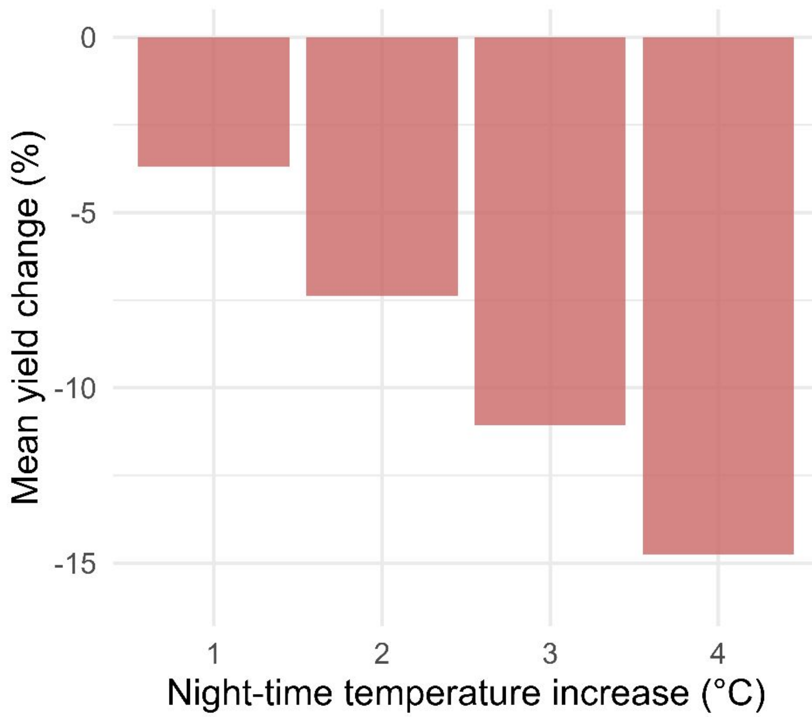

Figure 2. Mean maize yield change (%) in simulated scenarios with increasing NT during the June-July period in the Mid-South US. Red bars represent the average yield change across all scenarios for each of 1°C.

4. DISCUSSION:

Our key findings were the following: a) MinT (a proxy for night-time temperature) was the main environmental variable driving maize yield in the Mid-South US (β = -0.54), followed by CumRad, and rain during May and b) for every 1°C increase in NT, maize yield decreased 3.7% which corresponds to an average of 500 kg ha^{-1} relative to each location x hybrid x year baseline yield. Given an average yield of 13730 kg ha^{-1}, the RMSE represents 8%, and the cross-validated error corresponds to 11%.

Previous research has shown that high NT reduces yield mainly by decreasing kernel number (Kettler et al., 2022, 2024; Wang et al., 2020). Yield penalty is primarily driven by increased respiration rates and by impaired reproductive processes. Our results are consistent with previous studies that reported a 14% yield penalty when the NT was increased by 3.8°C which represents a 3.68% decrease for every 1°C change (Hein et al., 2024). Similarly, Wang et al. (2020) reported a 23.8% yield loss and a 25.1% kernel number loss when the NT was increased by 8°C mainly driven by reduced pollen viability and increased respiration rates. Accelerated development rates and elevated transpiration reduce the carbon availability for kernel set, diminishing photosynthates produced during the day. As a result, the grain filling period can be shortened by 5 days which can cause an 8% decrease in kernel number and deplete kernel weight (Cantarero et al., 1999).

Study limitations exist. The overall performance of the model was good, but the increase in error under cross-validation compared to the RMSE reflects the challenge of predicting yields at new locations. However, models using environmental covariates typically improve prediction accuracy for untested sites by decreasing out-of-location prediction error (Buntaran et al., 2021; Tadese et al., 2024). Future studies should focus on management practices to mitigate the negative effect of NT on maize yields and to evaluate whether hybrids of different relative maturity respond differently to increases in NT.

5. CONCLUSION:

Our study generates new insights into the impacts of high night-time temperatures on maize yield, an emerging yet underappreciated environmental threat to US maize production.

Our mixed-effect model approach allowed us to quantify the direction and magnitude of the impact that night-time temperature had on maize final grain yield in the Mid-South US. For every 1°C increase in night-time temperature, yields decreased 3.7% (500 kg ha^{-1}). While the results are specific to our model structure and setup, the conclusions can be extrapolated to maize production systems in additional environments, as the model predictors are mostly environmental covariates. The findings of the study can help refine management practices, such as planting date and hybrid relative maturity selection, to minimize the yield penalty caused by high NT, and can help enhance maize yield predictions using NT as a predictor variable in statistical models.

ACKNOWLEDGMENTS

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

ORCID

Caio Dos Santos: 0000-0003-0766-6764

Jason Kelley: 0009-0005-4588-4203

Elvis Felipe Elli: 0000-0001-9247-4956

SUPPLEMENTAL MATERIALS

Supplemental material is included in this paper. The file includes additional figures and table for a deeper site and weather description, yield variability summaries across the entire data set and model results. The material provides additional information of the environments to the readers, supports and expands the findings of the study.

AUTHOR CONTRIBUTIONS

Maximo Nores Allende: Conceptualization; data curation; formal analysis; software; visualization; writing original draft.

Elvis F. Elli: Conceptualization, investigation; resources, validation; writing—review and editing.

Caio dos Santos: Validation; data analysis, writing—review and editing

Jason Kelley: writing—review and editing, methodology.

John Carlyn: Investigation; methodology; resources, writing, review, and editing.

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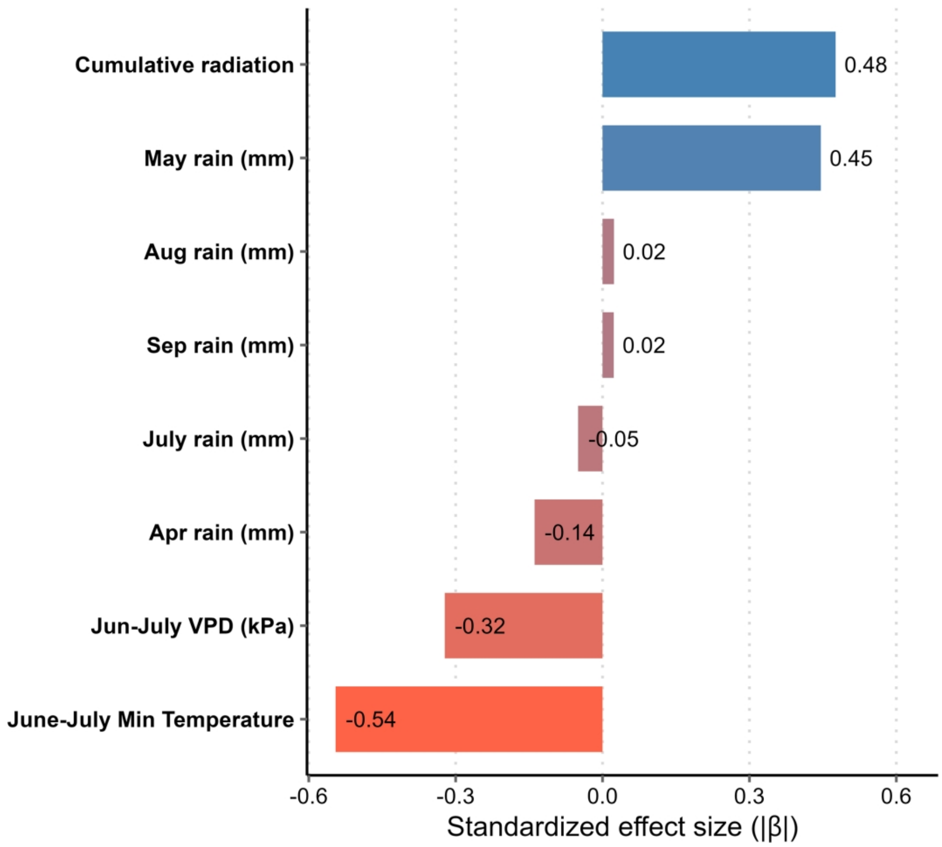

Figure 1. Relative importance of standardized climate predictors on determining maize yield across 64 site-years of data. Bars show standardized effect sizes with positive predictors in blue and negative in red.

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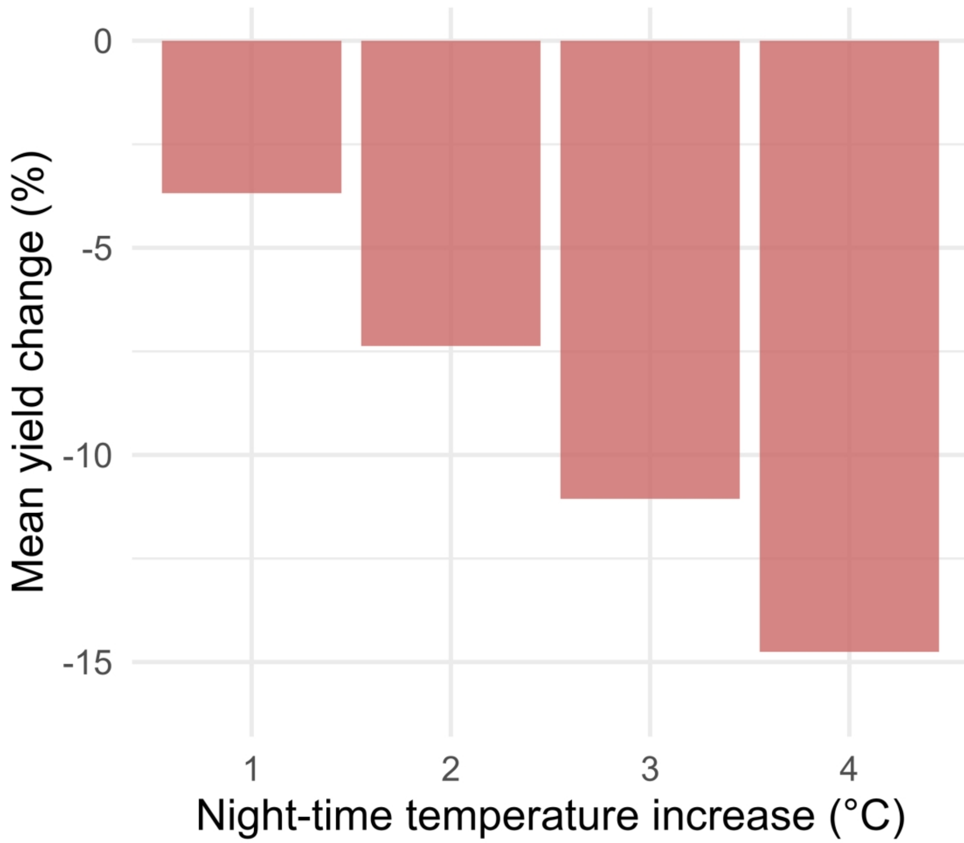

Figure 2. Mean maize yield change (%) in simulated scenarios with increasing NT during the June-July period in the Mid-South US. Red bars represent the average yield change across all scenarios for each of 1°C.

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Supplementary Material

Night-time temperatures drive corn yield variability in the US Mid-South

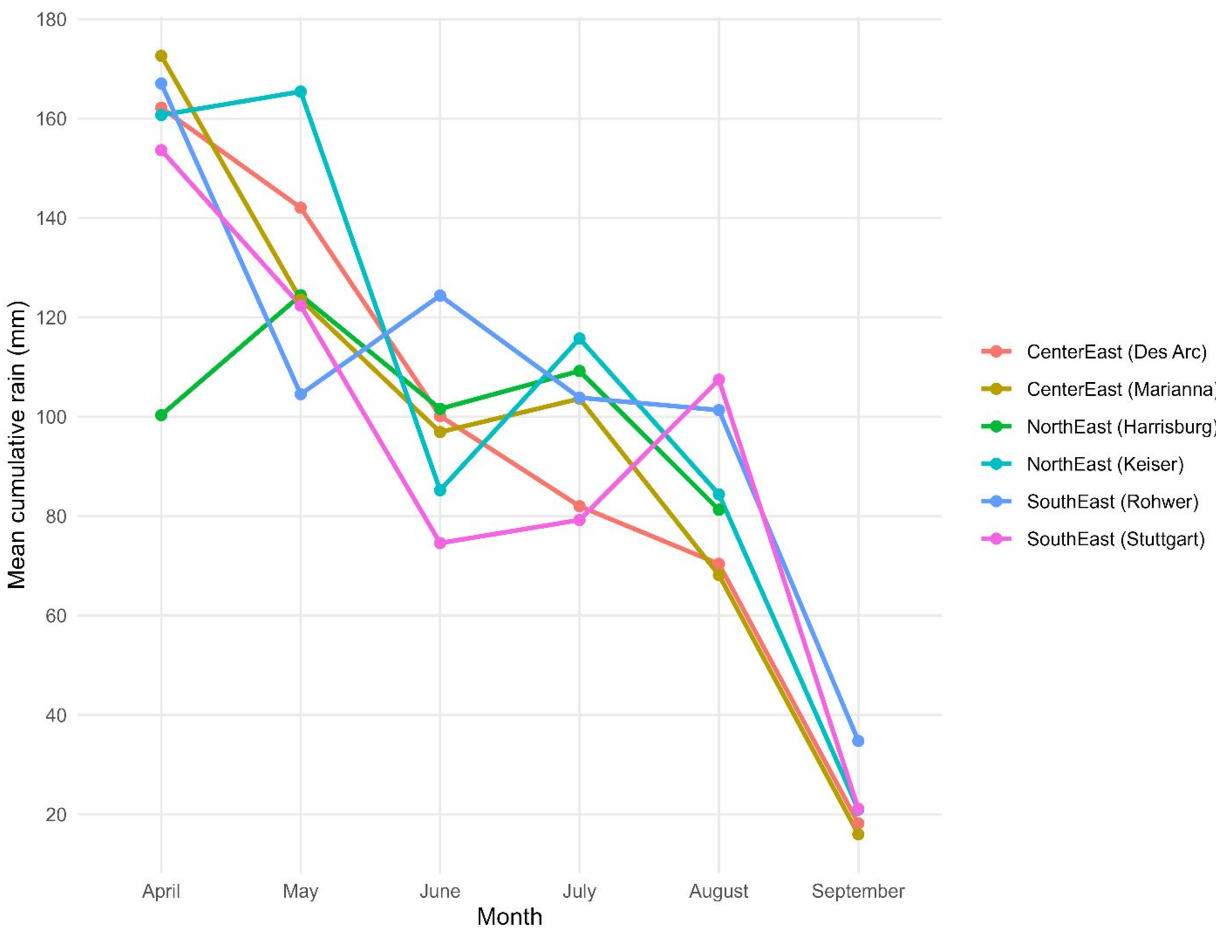

Figure S1. Cumulative monthly rainfall across moths and locations in Eastern Arkansas. Lines and points represent the average cumulative rain for each month and colors indicating locations.

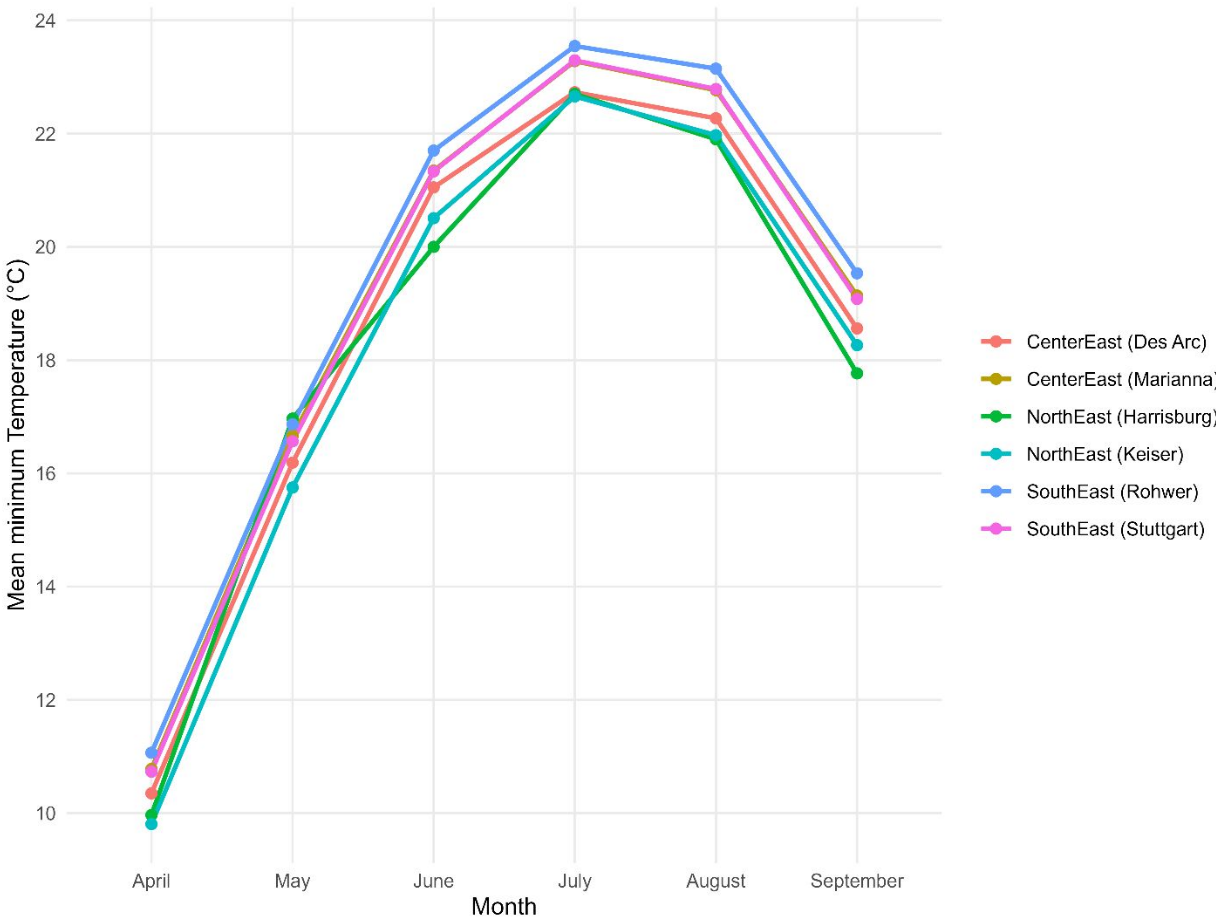

Figure S2. Minimum temperature across months and locations in Eastern Arkansas. Lines and points represent the average minimum temperature for each month.

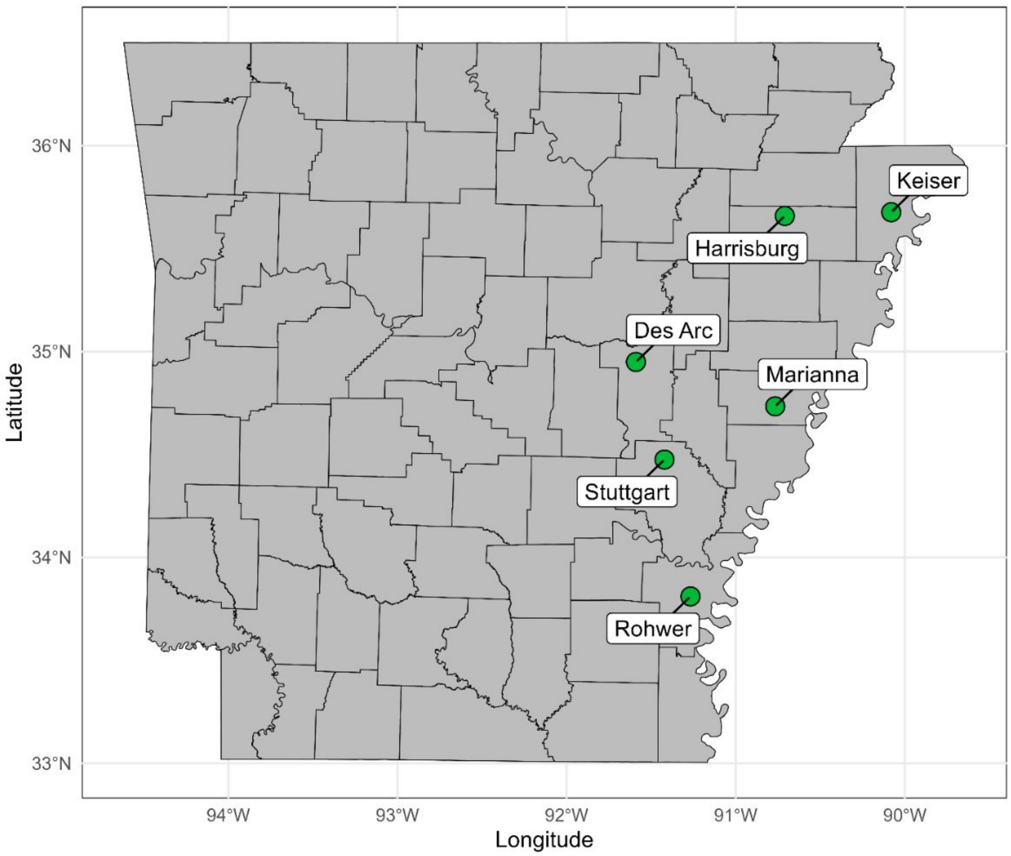

Figure S3. Locations of experimental fields (green pins) in eastern Arkansas where variety testing data were collected from 2010 to 2024.

Table S1. Summary of yield variability by location across 15 years.

Location	Years (n)	Mean Yield (kg ha^{-1})	Minimum Yield (kg ha^{-1})	Maximum Yield (kg ha^{-1})	Range (kg ha^{-1})
Harrisburg	3	15400	10800	18000	7200
Des Arc	10	13470	9700	17700	8000
Stuttgart	12	13960	8600	18000	9400
Marianna	12	13450	8500	19700	11200
Keiser	13	13570	9000	18200	9200
Rohwer	14	13920	7800	18700	10900

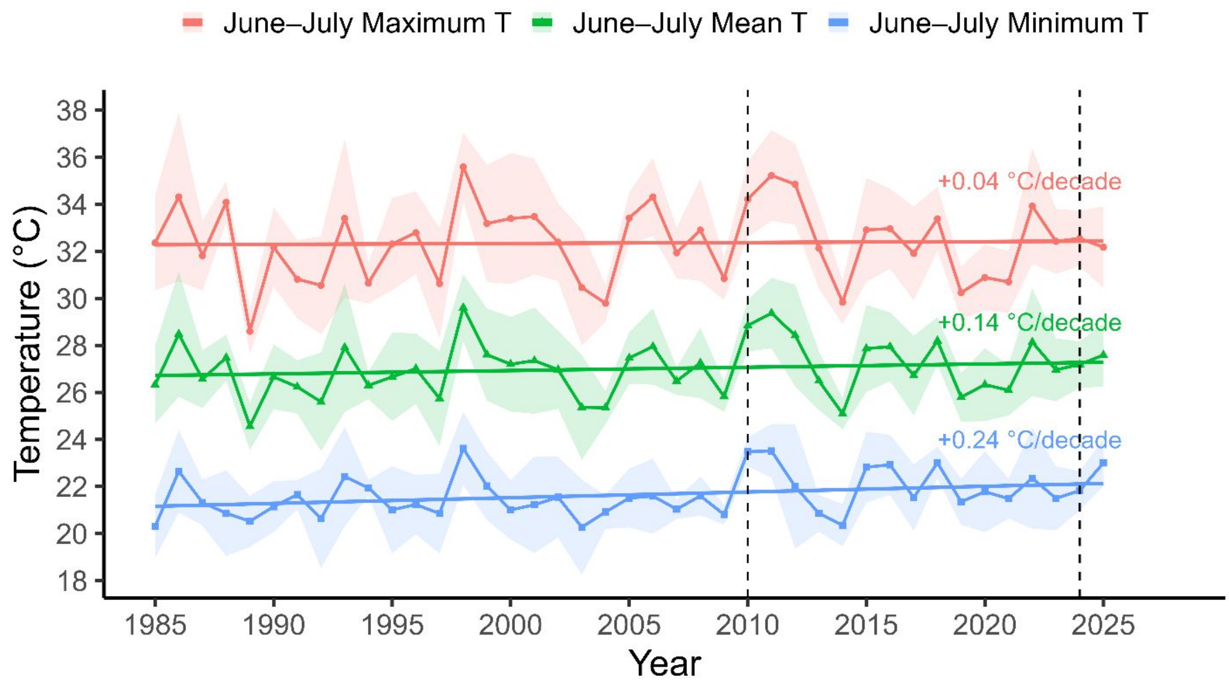

Figure S4. June-July temperature across 40 years of data. Colors represent maximum, average and minimum temperature during the June-July period for the last 40 years in Arkansas with lines representing the long-term trend and ribbons showing the within-year variability. Vertical dashed lines represent the time period during which the experiments were conducted.

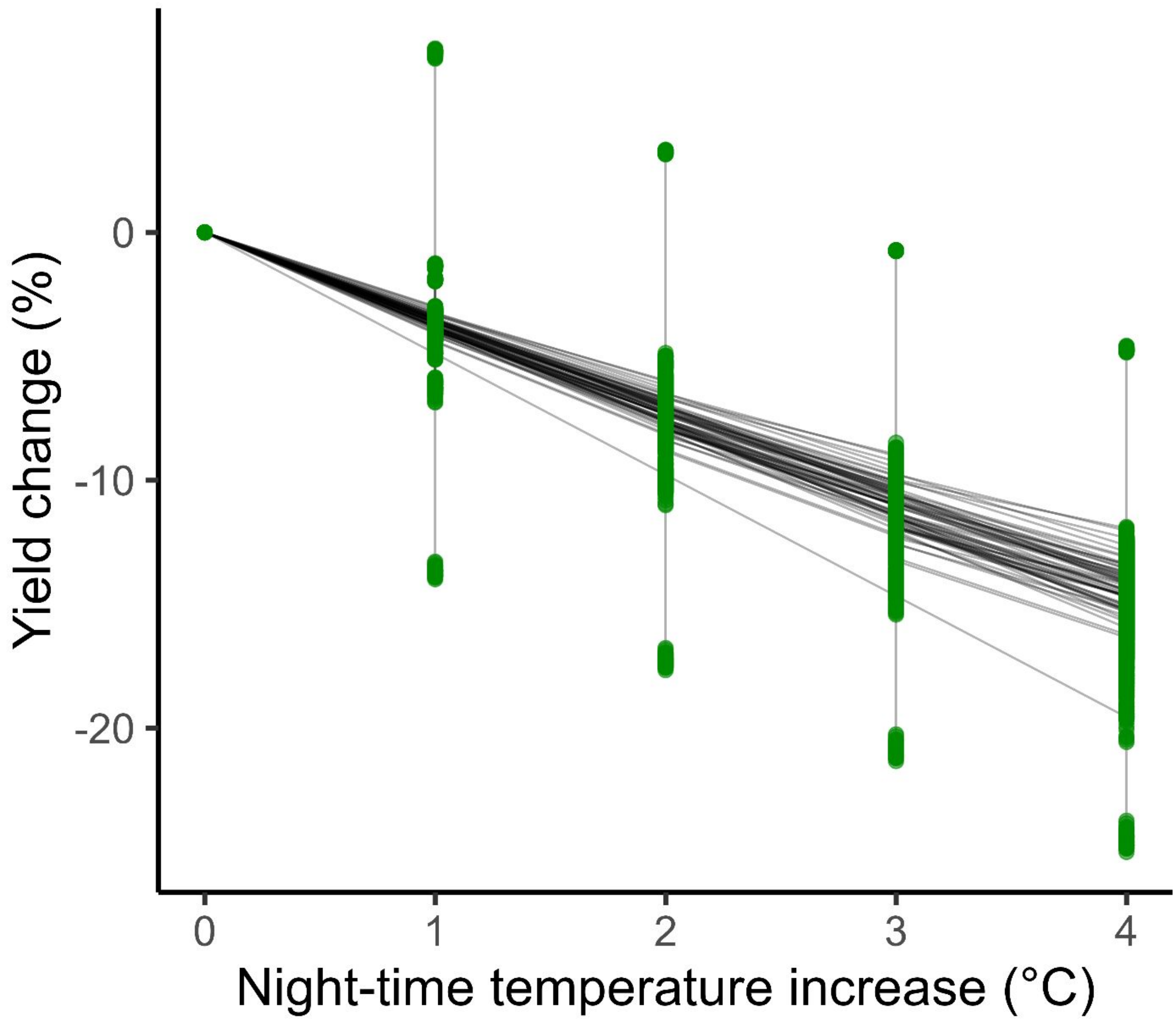

Figure S5. Effect of increasing night-time temperature on corn yield: Percentage change in yield under simulated scenarios with increasing Night-time temperature relative to the baseline scenario (model predictions under observed weather conditions).